

```

1  #include <stdio.h>
2
3  int main() {
4      int nof, nor;
5
6      printf("Enter number of frames: ");
7      scanf("%d", &nof);
8
9      printf("Enter number of pages: ");
10     scanf("%d", &nor);
11
12     int ref[50];
13
14     printf("Enter reference string:\n");
15
16     for (int i = 0; i < nor; i++) scanf("%d", &ref[i]);
17
18     int frame[10];
19
20     for (int i = 0; i < nof; i++) frame[i] = -1;
21
22     int pageFaults = 0;
23
24     printf("\nPage\tFrames\n");
25
26     for (int i = 0; i < nor; i++) {
27         int found = 0;
28         // checking for page hit
29         for (int j = 0; j < nof; j++) {
30             if (frame[j] == ref[i]) {
31                 found = 1;
32                 break;
33             }
34         }
35
36         if (!found) {
37             int pos = -1;
38             // checking for empty frame
39             for (int j = 0; j < nof; j++) {
40                 if (frame[j] == -1) {
41                     pos = j;
42                     break;
43                 }
44             }
45             // Need replacement
46             if (pos == -1) {
47                 int lruIndex = i;
48                 pos = 0;
49
50                 for (int j = 0; j < nof; j++) {
51                     int k;
52
53                     for (k = i - 1; k >= 0; k--) {
54                         if (frame[j] == ref[k]) break;
55                     }
56
57                     if (k < lruIndex) {
58                         lruIndex = k;
59                         pos = j;
60                     }
61                 }
62             }
63
64             frame[pos] = ref[i];
65             pageFaults++;
66         }
67
68         printf("%d\t", ref[i]);
69         for (int j = 0; j < nof; j++) printf("%d ", frame[j]);
70         printf("\n");
71     }
72     printf("\nTotal Page Faults = %d\n", pageFaults);
73 }

```